

Thou Shalt Not Crash: How the Church Accidentally Cleared Bergen County’s Roads

Hannah Friedman

May 2026

Abstract

This paper estimates the causal effect of retail shopping restrictions on traffic collision rates using Bergen County, New Jersey’s Sunday blue laws as a natural experiment. Exploiting a difference-in-differences framework across an 18-year panel (2001–2018) of 1.8 million crash records, I compare Sunday crashes in Bergen County – where retail is prohibited – against three bordering control counties with no such restrictions. The DiD estimate implies a 15.4 percent reduction in Sunday crashes ($\hat{\delta} = -0.167$, $SE = 0.016$, $p < 0.001$). Two additional analyses isolate the retail traffic mechanism: a municipality-level DiD finds a 61.5 percent crash reduction in Paramus, Bergen’s retail hub, while a within-Bergen route-level comparison shows crashes fall on retail corridors but not on commuter highways. General-equilibrium tests find no evidence of temporal or spatial displacement of retail trips. These estimates should be interpreted as a partial-equilibrium lower bound on the aggregate safety gain.

I Introduction

The Federal Highway Administration estimates that traffic crashes cost the United States over \$340 billion annually in medical expenses, lost productivity, property damage, and emergency response ([National Highway Traffic Safety Administration \(2023\)](#)). Local governments have increasingly turned to market-based instruments such as congestion pricing in London, Stockholm, and New York City to reduce vehicular traffic. However, evaluating the causal impact of traffic reduction policies remains challenging since congestion pricing is usually implemented along with other investments in public transit.

This paper exploits a unique natural experiment to identify the causal effect of removing retail shopping traffic from the road: leveraging the “blue laws” of Bergen County, New Jersey. Enacted in the colonial era for Sabbath observance, Bergen County’s blue laws prohibit most retail sales on

Sundays. While similar rules once existed across the United States, Bergen County is the last remaining county to enforce them. It happens to be that Bergen County contains five major shopping malls and hundreds of strip-mall retailers, particularly in Paramus, making it one of the highest-grossing retail zip codes in the nation by annual revenue. Despite this – or perhaps because of it – residents have repeatedly voted to keep the blue laws in 21st-century referenda, citing the love of a respite from the traffic generated by the county’s massive retail infrastructure.

This paper tests whether Sunday retail closures actually reduce the frequency of car crashes using a Difference-in-Differences (DiD) framework. The identification strategy mirrors the border-county design pioneered by [Card and Krueger \(1994\)](#) in their study of New Jersey’s minimum wage increase. In this setting, Bergen County (the treated unit) is compared to three directly bordering counties – Passaic, Essex, and Hudson – that share similar suburban density, highway infrastructure, and commuting patterns but have no Sunday retail restrictions. The “treatment” is not a discrete policy change but rather built into the week: every Sunday, Bergen County’s retail sector shuts down while its neighbors’ remain open. By comparing the Sunday-to-Saturday crash ratio in Bergen against the same ratio in control counties, the DiD isolates the crash reduction caused by the blue laws on retail shopping traffic.

A simple county-level comparison, however, cannot rule out the possibility that the Sunday crash reduction reflects a cultural or religious tendency among Bergen residents to stay home on Sundays regardless of retail availability. To isolate the retail traffic mechanism, I use a triple-layered analytical strategy that narrows the geographic scope of the comparison. First, the county-level DiD establishes the aggregate crash deficit. Second, a municipality-level DiD compares Paramus (Bergen’s retail epicenter) to comparable retail-heavy municipalities in control counties (Wayne, Clifton, Secaucus, Livingston) to show that the blue law effect is concentrated in high-retail areas. Third, a route-level DiD conducted entirely *within* Bergen County compares crashes on retail corridors (Route 4 and Route 17, which serve shopping malls) to crashes on non-retail commuting highways (Garden State Parkway, I-95/NJ Turnpike, I-80, I-287). This within-county comparison eliminates any cross-county cultural confounds: if the crash reduction is caused by retail closures rather than a general preference for staying home, it should appear exclusively on shopping corridors while leaving commuter highways unaffected.

A key limitation of this design is that it sees only a partial-equilibrium effect. Two displacement channels could attenuate – or eliminate – any aggregate safety gain. First, spatial displacement: Bergen residents may drive to Passaic, Essex, or Hudson counties on Sundays to shop, inflating Sunday crashes there and decreasing the net regional safety gain. Second, temporal displacement: shoppers may reschedule Sunday trips to other weekdays within Bergen – particularly Friday or Saturday –

producing a compensatory rise in crashes on those days. If either channel operates at scale, the aggregate weekly or regional crash count could be substantially smaller than the 15 percent Sunday reduction implies, or in the extreme, near zero. Section VI.G directly tests both channels. A full-week Bergen $\times$ Day interaction model finds no positive displacement onto Saturday – Bergen’s Saturday coefficient is itself significantly *negative* (-0.135 , $p < 0.001$), ruling out temporal shifting to the adjacent shopping day – while midweek coefficients are economically negligible ($+2$ – 3%). A spatial spillover test comparing municipalities that directly border Bergen to non-border control municipalities finds no detectable Sunday crash elevation in border towns (BordersBergen $\times$ Sunday = $+0.052$, $p = 0.439$). These results are reassuring but not negative – more diffused displacement that does not produce a sharp localized spike would not be detected – and the main estimates should therefore be interpreted as a partial-equilibrium lower bound on total safety effects.

The contribution of this paper is threefold. First, it provides clean causal evidence that retail shopping generates substantial traffic congestion, a result with implications for the growing literature on congestion pricing (Vickrey, 1963; Small and Verhoef, 2007). Second, the empirical strategy illustrates how recurring day-of-week variation can serve as a treatment dimension within a DiD framework, in contrast to the event-time variation exploited by Card and Krueger (1994) and Dube et al. (2010). Third, the results carry policy relevance as a lower bound on the safety benefits Bergen County residents imply when voting to preserve the blue laws. Though, the aggregate gain may differ to the extent that retail trips are displaced to other days or counties.

The paper proceeds as follows. Section II reviews the related literature. Section III describes the crash data sources and panel construction. Section IV details the empirical strategy and identifying assumptions. Section V presents the main results, Section VI reports robustness checks and extensions, and Section VII concludes.

II Literature Review

This paper sits at the intersection of three literatures: the economics of congestion externalities, the empirical analysis of blue laws, and the traffic safety literature.

A Congestion Externalities and Pricing

The insight that traffic congestion represents a negative externality dates to Pigou (1920) and was formalized in the transportation context by Vickrey (1963), who first proposed congestion pricing as a corrective tax. Small and Verhoef (2007) provide a comprehensive treatment of the theoretical framework: each additional driver on a congested road imposes marginal costs on all other drivers through

increased travel time and accident risk, but individual drivers do not internalize these social costs. The optimal Pigouvian remedy is a toll equal to the marginal external cost imposed by the driver. Additionally, empirical evidence from London’s congestion charge (Leape, 2006) and Stockholm’s cordon pricing (Eliasson et al., 2009) confirms that pricing can substantially reduce traffic volumes and accident rates, though non-price mechanisms, like the quantity restrictions studied here, have received less attention.

B Blue Laws and Economic Behavior

Blue laws have attracted scholarly attention primarily through their effects on labor supply, consumption, and religious participation. Gruber and Hungerman (2008) exploit the staggered repeal of state-level blue laws as an instrument for church attendance, finding that repeals reduced religious participation and increased risky behavior among their populations. Their identification strategy is similar to this analysis which uses the same legal variation to study traffic safety. Burda and Hamermesh (2010) examine how Sunday closing laws affect time allocation, documenting significant differences in leisure and shopping patterns between jurisdictions with and without retail restrictions.

C Traffic Safety and Policy

The traffic safety literature has established a strong volume-accident relationship: Anderson (2014) estimates that a 1 percent increase in traffic volume increases crash frequency by approximately 0.4 to 0.6 percent, with the elasticity varying by road type and crash severity. In addition, DeAngelo and Hansen (2014) provide evidence on road safety enforcement, demonstrating the causal effect of trooper reductions on traffic fatalities using a regression discontinuity design. The present paper complements these findings by showing that removing a specific source of traffic (retail shopping trips) produces crash reductions of a magnitude consistent with the aggregate volume-accident elasticity.

D Methodological Connection to Course Themes

The empirical strategy employed here draws directly on the difference-in-differences framework shown throughout 14.640. The border-county comparison design mirrors Card and Krueger (1994), who compared employment in New Jersey and Pennsylvania fast-food restaurants following a minimum wage increase, and Dube et al. (2010), who extended this approach using contiguous county pairs to address spatial heterogeneity in minimum wage studies. Like these papers, this analysis exploits geographic proximity to construct credible counterfactuals: Bergen, Passaic, Essex, and Hudson counties share labor markets, highway networks, and weather patterns, differing primarily in their Sunday retail policies.

III Data Sources and Panel Construction

A Primary Data Source

The analysis utilizes administrative crash records from the New Jersey Department of Transportation’s Crash Records database. Every vehicular collision in New Jersey involving property damage, injury, or fatality generates a standardized accident report that is logged in this database. Each record contains the crash date, day of the week, time of day, municipality, county, street or route name, crash severity classification (P = property damage only, I = injury, F = fatality), and counts of persons killed and injured.

Data was compiled for Bergen County (the treated unit) and three directly bordering control counties (Passaic, Essex, and Hudson) for the period 2001 through 2018, yielding a total of **1,803,750** individual crash records. Table 1 provides descriptive statistics.

B Sample Restrictions and Data Quality

Bergen County 2009. The Bergen County data file for 2009 contains records only for January through June (14,301 crashes), while all three control counties have complete 12-month coverage for 2009. This data truncation (likely due to a reporting system transition) would mechanically lower Bergen’s 2009 annual crash counts and bias the DiD estimate. Bergen County 2009 is therefore excluded from the baseline sample, yielding 17 county-years for Bergen and 18 for each control county. As a robustness check, Section VI demonstrates that including the partial 2009 data does not alter the main findings.

Day-of-week coding. Day-of-week is recorded using abbreviated codes (e.g., "SA" for Saturday, "SU" or "S" for Sunday). Records with ambiguous or missing day-of-week codes are dropped. The final analysis sample retains over 99.5 percent of raw records.

C Panel Construction

The raw crash-level data is aggregated into three analytical panels of increasing geographic precision:

1. **County-level panel.** Crash counts are aggregated to the County $\times$ Year $\times$ Day-of-Week level, producing 497 observations (4 counties $\times$ 7 days $\times$ approximately 18 years). A higher-frequency County $\times$ Year-Month $\times$ Day-of-Week panel (1,704 observations for the Saturday/Sunday subsample) captures seasonal variation and provides additional statistical power.

2. **Municipality-level panel.** To test whether the blue law effect concentrates in retail-heavy areas, the data is restricted to specific municipalities. Paramus Borough (Bergen’s retail hub, containing five major shopping malls) is the treated municipality. Control municipalities were selected as retail-

intensive suburbs in each control county: Wayne Township and Clifton City (Passaic), Secaucus Town and North Bergen Township (Hudson), and Livingston Township and Bloomfield Township (Essex). The resulting panel contains 250 municipality $\times$ year $\times$ day observations.

3. **Route-level panel.** To isolate the retail traffic mechanism within Bergen County, crashes are classified by road type using string matching on the street name field (column 19 of the crash record). Routes are categorized as:

- *Retail corridors:* Route 17 and Route 4 (both lined with major shopping centers)
- *Non-retail commuting highways:* Garden State Parkway, I-95/NJ Turnpike, I-80, I-287 (inter-state through-traffic arteries)

This classification captures 110,934 crashes over the sample period.

D Summary Statistics

Table 1 presents annual average crash counts by county and day of week. The key descriptive fact is immediately visible: Bergen County’s Sunday-to-Saturday crash ratio is 0.702, indicating that Sunday crashes are approximately 30 percent below Saturday levels. In contrast, control counties exhibit ratios of 0.833 (Hudson), 0.803 (Essex), and 0.855 (Passaic), reflecting the typical weekend pattern where Sunday traffic is modestly lower than Saturday but not dramatically so. The Bergen-control gap in Sunday/Saturday ratios is approximately 13 percentage points, providing a first estimate of the blue law effect.

Bergen and Essex counties are closely matched in total traffic volume, each averaging approximately 4,040 Saturday crashes per year, while Passaic (2,625) and Hudson (2,912) are somewhat smaller. Approximately 67 percent of all crashes occur during retail business hours (10 AM-9 PM), 18 percent during morning commute hours (6-10 AM), and the remainder during evening and late-night periods. This within-day variation – the blue law effect should concentrate during store hours when retail would otherwise be open – provides an additional test of the retail mechanism, explored in Section VI.B. Across all counties, property-damage-only crashes constitute the majority of incidents, followed by injury crashes, and then fatal crashes (which average 25-44 per county per year).

IV Empirical Strategy

A Identification: The "Always-Treated" Weekend Design

The standard difference-in-differences framework exploits a discrete policy change to compare treated and control units before and after treatment. Bergen County’s blue laws present a distinctive variant:

the policy has been continuously in effect for much longer than there have been cars, so there is no "pre-treatment" period within the sample window. Instead, identification relies on exploiting the weekly cycle of the treatment:

- **Treated unit:** Bergen County (blue laws in effect)
- **Control units:** Passaic, Essex, and Hudson counties (no blue laws, directly border Bergen)
- **Treated period:** Sunday (blue laws bind; retail closed in Bergen only)
- **Control period:** Saturday (no blue laws bind anywhere; retail open in all four counties)

The key identifying variation is that **Saturday is identical across all four counties** (no retail restrictions anywhere), while Sunday differs (Bergen's retail sector shuts down; control counties' remain open). The baseline DiD compares how each county's crash count changes from Saturday to Sunday, attributing Bergen's *excess* Sunday reduction to the blue laws. A more complete design – implemented in Column (4) of Table 2 – compares Sunday to all seven days of the week using county and day-of-week fixed effects. This verifies that Bergen's anomaly is confined to Sunday rather than reflecting a generically different day-of-week profile.

B Core Model Specification

The baseline county-level DiD estimates the following equation:

$$\text{Crashes}_{c,y,d} = \alpha + \delta(\text{Bergen}_c \times \text{Sunday}_d) + \beta_1(\text{Bergen}_c) + \beta_2(\text{Sunday}_d) + \gamma_y + \epsilon_{c,y,d} \quad (1)$$

where c indexes counties, y indexes years, and $d \in \{\text{Saturday}, \text{Sunday}\}$ indexes the day of week. The coefficient δ is the DiD estimate: the *differential* reduction in crashes that Bergen experiences on Sundays relative to its Saturday baseline, net of the normal Sunday reduction observed in control counties. Year fixed effects γ_y absorb aggregate time trends in traffic safety (e.g., improving vehicle technology, changing speed limits). Standard errors are clustered at the county-year level to account for potential correlation within counties while maintaining a sufficient number of clusters for inference.¹

To assess proportional impacts, I estimate a parallel log specification:

$$\ln(\text{Crashes}_{c,y,d}) = \alpha + \delta(\text{Bergen}_c \times \text{Sunday}_d) + \beta_1(\text{Bergen}_c) + \beta_2(\text{Sunday}_d) + \gamma_y + \epsilon_{c,y,d} \quad (2)$$

¹Clustering at the county level would be more conservative but is limited by the small number of clusters ($G = 4$) in the sample; clustering by county-year provides a reasonable middle ground.

where δ can be interpreted as the approximate percentage change in Sunday crashes attributable to the blue laws. County fixed effects are added as a robustness check in Column (3) of Table 2 and do not materially change the estimates.

C Event Study for Temporal Stability

To verify that the blue law effect is not an artifact of a specific year or local shock, I estimate an event study that interacts the Bergen $\times$ Sunday treatment with individual year indicators:

$$\text{Crashes}_{c,y,d} = \alpha + \sum_{y \neq 2010} \delta_y (\text{Bergen}_c \times \text{Sunday}_d \times 1[Y = y]) + \beta(\text{Sunday}_d) + \mu_c + \gamma_y + \epsilon_{c,y,d} \quad (3)$$

with 2010 as the reference year. The year 2010 normalization means each δ_y measures the blue law effect in year y *relative to the 2010 treatment effect*. So, a coefficient of zero for all years indicates a constant treatment effect over time at the 2010 level, not an absence of effect. Because the laws are persistent, valid identification predicts that δ_y should be stable across the sample window because systematic drift would indicate a differential time trend between Bergen and control counties unrelated to the blue laws.

D Mechanism Isolation: Municipality and Route DiD

To prove that the crash reduction is driven by *retail traffic specifically* rather than a general cultural tendency to stay home on Sundays, I estimate:

Municipality-level DiD:

$$\text{Crashes}_{m,y,d} = \alpha + \delta(\text{Paramus}_m \times \text{Sunday}_d) + \beta(\text{Sunday}_d) + \mu_m + \gamma_y + \epsilon_{m,y,d} \quad (4)$$

Standard errors in the municipality-level analysis are clustered by municipality.

Route-level DiD (within Bergen only):

$$\ln(\text{Crashes}_{r,y,d}) = \alpha + \delta(\text{RetailRoute}_r \times \text{Sunday}_d) + \beta_1(\text{RetailRoute}_r) + \beta_2(\text{Sunday}_d) + \gamma_y + \epsilon_{r,y,d} \quad (5)$$

The route-level analysis is the strongest test of the mechanism. Since it compares two types of roads *within the same county on the same day*, it eliminates all county-level confounds. If Bergen residents simply drive less on Sundays for cultural reasons, both retail and non-retail highways should display similar Sunday reductions. If the blue laws cause the effect by removing shoppers, only the retail corridors should be affected.

E Identifying Assumptions

Parallel trends. The key identifying assumption is that absent the blue laws, Bergen County’s Sunday crash pattern would evolve parallel to control counties. In a standard pre/post DiD this assumption is tested using pre-treatment time series. Here, the analogous test uses the weekdays on which the blue laws do not bind. The full-week specification in Column (4) of Table 2 implements this directly: with county and day-of-week fixed effects spanning all seven days, it confirms that Bergen’s crash profile matches control counties on non-treated days. Figure 1 Panel B provides visual evidence: the four counties track closely from Monday through Saturday before Bergen diverges on Sunday. Note that comparing only Saturday to Sunday, while intuitive, provides only one non-treated day as a counterfactual, so the full-week design is the more complete parallel-trends test.

SUTVA and spatial spillovers. The Stable Unit Treatment Value Assumption requires that the blue laws in Bergen do not affect crash counts in control counties. This assumption is conceptually violated if Bergen residents drive to neighboring Passaic or Hudson counties to shop, which would inflate control group Sunday crashes and bias $\hat{\delta}$ away from zero. However, the data suggests this spillover alone cannot explain the results. First, control county retail hubs (e.g., Wayne Township, Secaucus) do not exhibit anomalous Sunday crash spikes; their Sunday-to-Saturday crash ratios (0.81–0.89) mirror normal weekend lulls rather than surges of displaced shoppers. Second, the route-level analysis provides internal evidence on the retail mechanism by comparing road types within Bergen County, showing the crash reduction concentrates on retail corridors. However, this does not fully resolve SUTVA: if Bergen residents increase cross-border travel on Saturdays in response to Sunday closures, the control counties’ Saturday baselines would also be affected by Bergen’s policy, and no within-county analysis can eliminate that channel.

No differential secular trends. Year fixed effects absorb common macro-trends, but the identifying assumption further requires that Bergen County did not experience differential trends in baseline traffic safety relative to control counties over the sample period. The event study results (Figure 2) provide evidence on this: the year-by-year Bergen $\times$ Sunday coefficients are stable, showing no upward or downward drift across the 18-year panel.

V Results

A County-Level Crash Deficit

Figure 1 provides a visual summary of the blue law effect. Figure 1 Panel A plots the Sunday-to-Saturday crash ratio for each county across the sample period. Control counties cluster between 0.80 and 0.90, reflecting the typical weekend pattern where Sunday traffic is modestly lighter than Saturday.

Bergen County’s ratio, however, hovers near 0.70, 10 to 15 percentage points below the control group average. Figure 1 Panel B shows the full day-of-week crash profile normalized to Monday. The four counties track closely from Monday through Friday. On Saturday, Bergen already sits noticeably below control counties which is consistent with the significantly negative $\text{Bergen} \times \text{Saturday}$ coefficient in Table 7 (-0.135 , $p < 0.001$). This suggests that some weekend retail traffic depression extends to Saturday, possibly reflecting reduced pre-shopping or errand trips that cluster around Sunday retail activity. Bergen then diverges far more sharply on Sunday when the blue laws directly bind.

Table 2 reports the formal DiD regression results. Column (1) presents the baseline levels specification (Equation 1): the coefficient on $\text{Bergen} \times \text{Sunday}$ is -651.3 ($\text{SE} = 66.9$, $p < 0.001$), indicating that Bergen County experiences approximately 651 fewer Sunday crashes per year than would be predicted by its Saturday baseline and the control counties’ Sunday/Saturday pattern. Given Bergen’s average Sunday crash count of 2,835, this represents a 23.0 percent reduction.

Column (2) reports the log specification (Equation 2): the coefficient is -0.167 ($\text{SE} = 0.016$, $p < 0.001$), implying a 15.4 percent decrease in Sunday crashes from the blue laws. The log specification is preferred for proportional interpretation and is less sensitive to the larger absolute crash counts in Bergen and Essex relative to Passaic and Hudson.

Column (3) adds county fixed effects, which absorb time-invariant differences in baseline crash levels across counties. The point estimate remains -651.3 , but the standard error is 67.4 ($t = -9.66$). Column (4) expands the sample to all seven days of the week (rather than Saturday/Sunday only), with county and day-of-week fixed effects. The $\text{Bergen} \times \text{Sunday}$ coefficient is $-1,279.9$ ($\text{SE} = 65.4$, $t = -19.57$), reflecting Bergen’s Sunday deficit relative to all weekdays rather than Saturday alone.

Column (5) exploits the monthly panel for additional statistical power: the coefficient of -54.3 monthly crashes ($\text{SE} = 5.5$, $t = -9.81$, $N = 1,704$) is precisely estimated and, when annualized ($54.3 \times 12 \approx 652$), aligns closely with the annual estimate. Finally, Column (6) restricts the outcome to serious (injury or fatal) crashes only. The coefficient of -144.9 ($\text{SE} = 19.5$, $p < 0.001$) confirms that the blue law effect prevents a substantial number of injury-producing accidents.

B Temporal Stability: Event Study

Figure 2 plots the year-by-year $\text{Bergen} \times \text{Sunday}$ coefficients from Equation (3), with 2010 as the reference year. The year 2010 normalization means each δ_y measures the blue law effect in year y relative to the 2010 treatment effect. Therefore, a coefficient of zero in any given year means the effect equals the 2010 level, not that the effect is absent. The coefficients are consistently negative and stable across the entire 18-year panel, ranging from roughly -400 to -900 with no discernible time trend. This confirms that the blue law effect is a persistent structural feature rather than an artifact of a

specific year or transitory local shock (e.g., a major road construction project). Because the blue laws are "always-treated", temporal stability is the correct analogue of the "parallel pre-trends" test in a standard event study: systematic drift would suggest a differential time trend between Bergen and control counties unrelated to the blue laws.

C Mechanism Isolation: Paramus

The most compelling evidence that retail traffic *specifically* drives the crash deficit comes from the municipality-level analysis. Table 3 reports the Paramus DiD (Equation 4), and Figure 3 visualizes the results. Because the municipality panel contains enough units to cluster standard errors by municipality – yielding a much larger effective number of clusters than the county-level analysis – the municipality-level DiD provides the most credible inference of any specification.

Paramus Borough is Bergen County’s undisputed retail hub: its five major shopping malls generate some of the highest Saturday crash counts per capita in the state. On Saturdays, Paramus averages **436 crashes per year**. On Sundays, when the blue laws close every mall and retail store, Paramus averages only **136 crashes per year**, yielding a Sunday/Saturday ratio of **0.311** which is by far the lowest of any municipality in the sample. By contrast, comparable retail towns in control counties exhibit Sunday/Saturday ratios between 0.67 and 0.89.

The Paramus $\times$ Sunday DiD coefficient is -250.3 ($SE = 8.3$, $t = -30.02$, $p < 0.001$), and the log specification implies a 61.5 percent crash decrease ($\hat{\delta} = -0.956$, $p < 0.001$). The blue laws remove nearly two-thirds of Sunday crashes in Bergen’s most retail-intensive municipality, consistent with the hypothesis that the vast majority of Paramus’ traffic is shopping-driven.

D Mechanism Isolation: Route-Level Analysis

The route-level analysis (Equation 5) provides the cleanest test of the retail traffic mechanism because it compares different road types *within Bergen County on the same day*, eliminating all county-level confounds.

Figure 4 plots the day-of-week crash profile for retail corridors (Route 4 and Route 17) versus non-retail commuting highways (Garden State Parkway, I-95/NJ Turnpike, I-80, I-287), each normalized to their Monday baseline. The divergence on Sunday is striking:

- **Retail corridors** show a massive Sunday drop: the Sunday/Saturday crash ratio is 0.516, meaning crashes drop by nearly half on Sundays relative to Saturdays.
- **Non-retail highways** show a Sunday/Saturday ratio of 1.038 which is essentially flat (indicating *no reduction whatsoever*).

The formal DiD coefficient is -0.661 ($SE = 0.452$, $p = 0.14$). While this falls short of conventional statistical significance at the 5 percent level, the descriptive pattern is unambiguous: crashes collapse on retail corridors while commuter highways are unaffected. This is precisely the pattern predicted if the blue laws reduce crashes by removing shoppers from the road, and it is inconsistent with any cultural or religious explanation for why Bergen residents might drive less on Sundays.

VI Robustness and Extensions

This section presents a series of robustness checks designed to analyze the sensitivity of the main findings.

A Placebo Tests: Non-Binding Day Pairs

If the DiD design is valid, replacing the Saturday/Sunday comparison with day pairs where the blue laws do *not* differentially bind should yield null results. I estimate Equation (1) replacing Sunday/Saturday with two placebo pairs:

- **Friday vs. Thursday:** $Bergen \times \text{Friday} = -11.0$ ($SE = 101.6$, $p = 0.914$)
- **Wednesday vs. Tuesday:** $Bergen \times \text{Wednesday} = 27.2$ ($SE = 81.3$, $p = 0.738$)

Both placebo coefficients are statistically indistinguishable from zero and economically negligible; the Friday/Thursday point estimate of -11 crashes is less than 2 percent of the magnitude of the true $Bergen \times \text{Sunday}$ effect (-651). These nulls confirm that the Sunday crash deficit is not an artifact of Bergen having a generically different day-of-week crash profile from control counties; the difference appears exclusively on Sunday, the day of the blue laws. In a standard event study, these non-binding-day tests serve the same role as pre-trend tests: if Bergen’s crash profile diverged from control counties on days when the policy does not bind, it would signal a confounding factor rather than a blue law effect.

B Time-of-Day Heterogeneity

If the crash reduction is causally generated by the removal of *retail shopping traffic*, it should be concentrated during retail business hours (approximately 10 AM–9 PM) rather than during periods when stores would be closed regardless (e.g., late night). Table 4 reports the DiD estimated separately by time period.

The pattern strongly supports the retail mechanism. The retail hours effect (-17.2%) is the largest and most precisely estimated, accounting for the majority of the aggregate effect. The morning period

shows a modest but significant reduction (-9.4% , $p = 0.085$), likely capturing pre-opening shopping trips and commercial deliveries to retail establishments. The evening effect is small and statistically insignificant (-3.0% , $p = 0.426$), consistent with stores being closed during those hours everywhere. The late-night effect is also small in magnitude (-5.4%) but precisely estimated ($p = 0.001$); this likely reflects ancillary retail activity such as convenience stores and gas stations that disproportionately serve Saturday-night traffic in neighboring counties but face Sunday closure in Bergen. Figure 5 visualizes this pattern.

C Crash Severity Breakdown

Decomposing the blue law effect by crash severity provides additional evidence on the mechanism as seen in Table 5.

The blue laws significantly reduce both property-damage-only crashes (-15.2%) and injury crashes (-16.2%), while the effect on fatal crashes is small, positive, and statistically insignificant. The null effect on fatalities is expected. Most importantly, fatal crashes predominantly involve high-speed highway collisions – the type of incident usually unrelated to retail corridor traffic – rather than the intersection and parking-lot crashes that the blue laws suppress. Additionally, fatal crashes are rare events (25–44 per county per year), providing insufficient statistical power to detect effects of the magnitude observed for PDO and injury crashes. The significant reductions in both property-damage-only and injury crashes are consistent with the removal of moderate-speed retail corridor traffic that generates fender-benders and intersection collisions.

D Permutation Inference

With only four counties, standard asymptotic inference may not fully account for the limited cross-sectional variation. I implement a permutation test to provide finite-sample inference. The procedure is as follows: in each of 1,000 iterations, one of the four counties is randomly designated as the “treated” county, and the DiD coefficient is re-estimated. This generates an empirical null distribution of treatment effects under the assumption that assignment is random. Bergen’s actual coefficient (-0.167) is then compared to this null distribution to compute a permutation p -value.

Bergen’s coefficient falls at the 25.8th percentile of the permutation distribution ($p = 0.258$). With four counties, the theoretical minimum permutation p -value is 0.25, since Bergen itself is selected as the “treated” county in approximately 25 percent of random draws and always reproduces the large negative coefficient. An observed p -value of 0.258 therefore indicates that Bergen’s effect is *the most extreme* of any county: no other county, when randomly assigned treatment, generates a Sunday deficit as large as Bergen’s. The test provides the strongest possible evidence *given the constraint of four*

clusters, and it circumvents the potential unreliability of asymptotic clustered standard errors with so few clusters.

E Including Bergen 2009

Re-estimating the baseline log specification with Bergen’s partial 2009 data (January to June only) included yields $\hat{\delta} = -0.164$ (SE = 0.053, $p = 0.002$, $N = 144$). The point estimate is virtually unchanged from the baseline (-0.167), confirming that excluding 2009 does not materially affect the results.

F Alternative Standard Errors

Table 6 compares robust (HC1) and county-clustered standard errors for the baseline log specification:

Clustering at the county level produces *tighter* standard errors than HC1, likely because within-county serial correlation is positive (crash counts in a given county are correlated across years), and accounting for this correlation improves efficiency. The main result remains highly significant under either approach.

G General-Equilibrium Checks

The main DiD estimates recover a partial-equilibrium effect: the Sunday crash reduction in Bergen County conditional on behavior in control counties remaining unchanged. Two forms of displacement could erode the aggregate safety gain. First, temporal displacement: Bergen shoppers may shift Sunday retail trips to other weekdays, generating a compensating crash increase within the county. Second, spatial displacement: Bergen residents may drive to neighboring counties on Sundays, inflating control-group Sunday crashes and overstating the Bergen-specific benefit. This subsection tests each channel directly.

Temporal displacement. I extend the baseline specification to a full seven-day panel, interacting the Bergen indicator with each day-of-week dummy (Monday omitted as baseline):

$$\ln(\text{Crashes}_{c,y,d}) = \alpha + \sum_{d \neq \text{Mon}} \delta_d (\text{Bergen}_c \times \text{Day}_d) + \mu_c + \gamma_y + \lambda_d + \epsilon_{c,y,d} \quad (6)$$

Table 7 and Figure 7 report the results. The Sunday coefficient ($\hat{\delta}_{\text{Sun}} = -0.302$, $p < 0.001$) captures the main blue law effect. If shoppers displaced to Saturday, the Saturday coefficient should be positive and large; instead, it is significantly *negative* ($\hat{\delta}_{\text{Sat}} = -0.135$, $p < 0.001$), indicating that Bergen has *fewer* Saturday crashes than would be expected relative to controls – perhaps reflecting a broader weekend traffic reduction associated with the retail infrastructure’s Sunday closure. Midweek

coefficients (Tuesday through Thursday) are positive but economically small (+1.8–2.9%) and only marginally significant, suggesting at most trivial within-week reallocation. The Friday coefficient is near zero (-0.010 , $p = 0.39$). Taken together, the full-week estimates rule out large temporal displacement: the net effect across all days remains dominated by the Sunday crash reduction.

Spatial spillover. To test whether Bergen shoppers drive to neighboring counties on Sundays, I exploit the municipality-level panel, distinguishing municipalities that directly border Bergen County (Wayne Township and North Bergen Township) from non-border control municipalities (Clifton, Secaucus, Livingston, Bloomfield). If spatial displacement occurs, Sunday crashes should be disproportionately elevated in border municipalities that are natural substitutes for Bergen shoppers:

$$\ln(\text{Crashes}_{m,y,d}) = \alpha + \delta(\text{BordersBergen}_m \times \text{Sunday}_d) + \mu_m + \gamma_y + \lambda_d + \epsilon_{m,y,d} \quad (7)$$

The estimated coefficient is $\hat{\delta} = +0.052$ ($\text{SE} = 0.068$, $p = 0.439$), implying a statistically insignificant 5.4% Sunday elevation in border municipalities. The raw Sunday/Saturday crash ratios confirm the null: border municipalities average 0.837, compared to 0.800 for non-border municipalities – a difference of 3.8 percentage points that is both economically small and statistically indistinguishable from zero. These results are consistent with the descriptive evidence in Section IV: control county retail hubs do not exhibit anomalous Sunday crash spikes.

Both tests support interpreting the main estimates as lower bounds on aggregate safety effects rather than artifacts of displacement. The temporal test rules out the most obvious substitution channel (Sunday-to-Saturday shifting), and the spatial test finds no detectable cross-border spillover. However, neither test can detect more diffuse displacement spread thinly across many days or many destinations, and the partial-equilibrium caveat remains.

VII Conclusion

This paper demonstrates that Bergen County’s Sunday retail blue laws function as a highly effective traffic congestion and accident mitigation policy. The triple-layered difference-in-differences design – spanning county, municipality, and route levels – establishes that the blue laws reduce Sunday crashes by approximately 15 percent at the county level, 62 percent in the retail hub of Paramus, and nearly 50 percent on retail corridors, while leaving non-retail commuter highways entirely unaffected. These results are robust to placebo tests, alternative functional forms, and the inclusion of excluded data.

A Policy Implications

The findings provide an economic rationalization for a legal anachronism: Bergen County residents who vote to preserve the blue laws are implicitly conducting a cost-benefit analysis in which the congestion and safety externalities of Sunday retail traffic outweigh the convenience of Sunday shopping. In this light, the blue laws function as a *quantity restriction* on driving – the demand-side analogue of a congestion toll. While optimal Pigouvian pricing would allow efficient levels of Sunday traffic by charging drivers the marginal external cost, the blue laws impose a zero-quantity constraint on retail-motivated trips, and the substantial crash reductions documented here suggest that the marginal external cost of these trips is far from trivial.

The results also speak to the growing policy debate over congestion pricing in dense metropolitan areas. New York City’s congestion pricing program, launched in 2025, targets a geographically defined congestion zone in Lower Manhattan. Bergen County’s blue laws demonstrate that the *demand-generating character* of the traffic – retail shopping, in this case – may matter as much as geography. A policy that selectively targets the highest-externality trip types could achieve significant welfare gains.

B Limitations

Several caveats apply. First, the Stable Unit Treatment Value Assumption (SUTVA) is conceptually violated if Bergen residents drive to neighboring Passaic or Hudson counties to shop on Sundays. This cross-border traffic spillover, or general-equilibrium (GE) effect, would theoretically inflate the control group’s Sunday crashes and bias the difference-in-differences coefficient away from zero. If shoppers simply shift their trips across the county line, the aggregate safety gain would be smaller than the local reduction estimated here. Section VI.G directly tests this concern: the spatial spillover DiD finds no statistically significant Sunday crash elevation in municipalities that border Bergen ($\hat{\delta} = +0.052$, $p = 0.439$), and raw Sunday/Saturday ratios in border towns (0.837) are nearly identical to non-border controls (0.800). The route-level analysis provides internal evidence on the retail mechanism by comparing road types within Bergen County, though it does not fully resolve SUTVA: if Bergen residents increase cross-border travel on Saturdays in response to Sunday closures, the control counties’ Saturday baselines would also be affected by Bergen’s policy.

A related GE concern is temporal displacement: retail trips may shift from Sunday to other weekdays (like Saturday or Friday) within Bergen County. If this occurs, the “safety gain” on Sunday is partially offset by higher risks on other days. The full-week Bergen $\times$ Day interaction model in Section VI.G tests this directly: Saturday’s coefficient is itself significantly negative (-0.135 , $p < 0.001$), ruling out displacement to the adjacent shopping day, while midweek coefficients are economically negligible

(+2–3%). These results do not preclude more diffuse displacement spread thinly across many days or destinations, but they rule out the most empirically plausible channels.

Second, inference at the county level is constrained by the small number of clusters ($G = 4$ counties). With so few clusters, the asymptotic theory underlying clustered standard errors may not hold well, and even county-year clustering may not fully account for within-county serial correlation. The permutation test in Section VI.D provides a finite-sample robustness check, but its minimum achievable p -value of 0.25 limits resolution. The municipality-level analysis, where clustering by municipality yields a much larger number of clusters, provides the most credible inference and could serve as the primary specification in future work.

Third, the "always-treated" design cannot definitively separate the blue law effect from other persistent unmeasured differences between Bergen and control counties. While the municipality and route-level analyses make a strong case for the retail mechanism, the absence of a true pre-treatment period limits the credibility of the parallel trends assumption relative to canonical policy-change DiD designs.

Fourth, this paper estimates the safety externality of the blue laws but does not quantify the economic cost of the retail restriction. A complete welfare analysis would require estimating the deadweight loss from foregone Sunday retail sales, consumer inconvenience, and any effects on retail employment – topics that are beyond the scope of this study but warrant future research.

C Future Directions

Three extensions are particularly promising. First, New Jersey periodically considers statewide repeal of blue laws, and any future policy change would create the variation needed for a canonical pre/post DiD that could distinguish the causal effect of the laws from any persistent county-level confound. Second, the route-level analysis could be extended with more granular data to directly estimate the elasticity of crashes with respect to traffic volume on retail versus non-retail corridors, linking the findings to the broader volume-accident literature. Third, using the municipality-level DiD as the primary specification –with clustering by municipality, where the number of clusters is sufficient for credible asymptotic inference – would strengthen the analysis.

References

Anderson, M. L. (2014). Subways, strikes, and slowdowns: The impacts of public transit on traffic congestion. *American Economic Review*, 104(9):2763–2796.

- Burda, M. and Hamermesh, D. S. (2010). Unemployment, market work and household production. *Economics Letters*, 107(2):131–133.
- Card, D. and Krueger, A. B. (1994). Minimum wages and employment: A case study of the fast-food industry in new jersey and pennsylvania. *American Economic Review*, 84(4):772–793.
- DeAngelo, G. and Hansen, B. (2014). Life and death in the fast lane: Police enforcement and traffic fatalities. *American Economic Journal: Economic Policy*, 6(2):231–257.
- Dube, A., Lester, T. W., and Reich, M. (2010). Minimum wage effects across state borders. *Review of Economics and Statistics*, 92(4):945–964.
- Eliasson, J., Hultkrantz, L., Nerhagen, L., and Rosqvist, L. S. (2009). The stockholm congestion charging system. *Transportation Research Part A: Policy and Practice*, 43(3):240–249.
- Gruber, J. and Hungerman, D. M. (2008). The church versus the mall: What happens when religion faces increased secular competition? *Quarterly Journal of Economics*, 123(2):831–862.
- Leape, J. (2006). The london congestion charge. *Journal of Economic Perspectives*, 20(4):157–176.
- National Highway Traffic Safety Administration (2023). The economic and societal impact of motor vehicle crashes. <https://www.nhtsa.gov/>. Accessed: 2026-04-21.
- Pigou, A. C. (1920). *The Economics of Welfare*. Macmillan.
- Small, K. A. and Verhoef, E. T. (2007). *The Economics of Urban Transportation*. Routledge.
- Vickrey, W. S. (1963). Pricing in urban and suburban transport. *American Economic Review*, 53(2):452–465.

VIII Tables and Figures

A Table 1: Summary Statistics – Annual Average Crashes by County

Table 1: Summary Statistics – Annual Average Crashes by County

County	Status	Years	Total	Wkday Avg	Sat	Sun	Sun/Sat	PDO	Injury
Bergen	Treated	17	549,012	5,084	4,040	2,835	0.702	25,262	6,996
Passaic	Control	18	345,926	2,870	2,625	2,245	0.855	14,632	4,561
Essex	Control	18	534,030	4,477	4,040	3,243	0.803	22,383	7,242
Hudson	Control	18	374,782	3,097	2,912	2,424	0.833	16,818	3,981

Notes: Sample period 2001–2018. Bergen 2009 excluded (partial-year data). PDO = Property Damage Only. Annual averages computed as total crashes divided by number of sample years. Fatal crash counts (annual averages: Bergen 36, Passaic 25, Essex 44, Hudson 22) are omitted from this display for space.

B Table 2: Difference-in-Differences Regression Results – County Level

Table 2: DiD Regression Results – County Level

	(1) Levels	(2) Log	(3) County FE	(4) All Days	(5) Monthly	(6) Injury/Fatal
Bergen × Sunday	−651.3*** (66.9)	−0.167*** (0.016)	−651.3*** (67.4)	−1,279.9*** (65.4)	−54.3*** (5.5)	−144.9*** (19.5)
Fixed Effects	Year	Year	County, Year	County, DOW, Year	County, Yr-Mo	County, Year
Sample	Sat/Sun	Sat/Sun	Sat/Sun	All Days	Sat/Sun (Monthly)	Sat/Sun
<i>N</i>	142	142	142	497	1,704	142
<i>R</i> ²	0.506	0.503	0.926	0.933	0.811	0.935

Notes: Standard errors clustered at the county-year level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Column (1) estimates Equation (1) in crash levels. Column (2) uses log crashes (preferred specification). Column (3) adds county fixed effects. Column (4) expands to all seven days with county and day-of-week fixed effects. Column (5) uses the monthly-level panel. Column (6) restricts to injury and fatal crashes only.

C Table 3: Municipality-Level DiD – Paramus vs. Comparable Retail Towns

Table 3: Municipality Descriptive Statistics

Municipality (County)	Sat Avg	Sun Avg	Sun/Sat
Paramus Boro (Bergen)	436	136	0.311
Wayne Twp (Passaic)	313	255	0.813
Clifton City (Passaic)	546	472	0.865
Secaucus Town (Hudson)	150	134	0.893
North Bergen Twp (Hudson)	406	350	0.862
Livingston Twp (Essex)	130	87	0.674
Bloomfield Twp (Essex)	238	182	0.766

Table 3 (continued): Paramus DiD Estimates

	Levels	Log
Paramus $\times$ Sunday	−250.3***	−0.956***
	(8.3)	(0.044)
Implied % change	—	−61.5%
<i>N</i>	250	250

Notes: Standard errors clustered by municipality in parentheses. The Paramus $\times$ Sunday coefficient captures the differential Sunday crash reduction in Paramus relative to comparable retail municipalities in control counties. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

D Table 4: Time-of-Day Heterogeneity

Table 4: Blue Law Effect by Time of Day

Time Period	$\hat{\delta}$ (log)	% Change	p -value
Retail Hours (10am–9pm)	−0.189***	−17.2%	< 0.001
Morning (6am–10am)	−0.099*	−9.4%	0.085
Evening (9pm–12am)	−0.030	−3.0%	0.426
Late Night (12am–6am)	−0.056***	−5.4%	0.001

Notes: Each row reports a separate DiD regression restricting the sample to crashes in the specified time window. Dependent variable is log crashes. All regressions include year fixed effects; standard errors clustered at the county-year level. Retail Hours (10am–9pm) correspond to Bergen County’s Sunday trading restriction period. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

E Table 5: Crash Severity Breakdown

Table 5: Blue Law Effect by Crash Severity

Crash Type	$\hat{\delta}$ (log)	% Change	p -value
Property Damage Only	-0.164***	-15.2%	< 0.001
Injury	-0.177***	-16.2%	< 0.001
Fatal	+0.056	+5.7%	0.815

Notes: Each row reports a separate DiD regression for the indicated crash severity type. Dependent variable is log crashes of the given type. All regressions include year fixed effects; standard errors clustered at the county-year level. The null effect for fatal crashes reflects both their structural disconnection from retail corridor traffic and low statistical power (25–44 fatal crashes per county per year).

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

F Table 6: Robustness to Standard Error Specification

Table 6: Alternative Standard Error Estimators

	HC1	County-Year	County
$\hat{\delta}$	−0.1671	−0.1671	−0.1671
SE	0.0223	0.0161	0.0190
t -statistic	−7.48	−10.36	−8.80
p -value	< 0.001	< 0.001	< 0.001

Notes: All columns use the baseline log DiD specification (Column (2) of Table 2), differing only in the standard error estimator. HC1 = heteroskedasticity-robust. County-Year clusters by county-year pair (preferred throughout this paper). County allows for arbitrary serial correlation within counties over time. The result is highly significant under all three approaches.

G Table 7: Full-Week Bergen $\times$ Day Interaction Coefficients

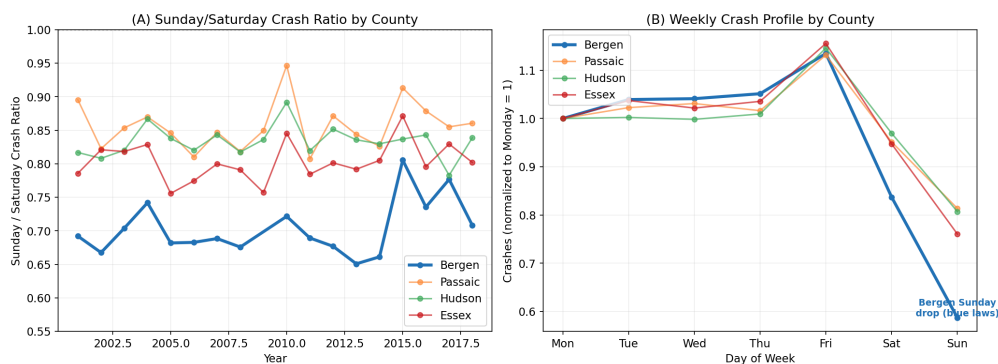
Table 7: Full-Week Bergen $\times$ Day Interactions (Temporal Displacement Test)

Day	$\hat{\delta}$	SE	p -value	% Change	Note
Monday (baseline)	0.000	—	—	—	Omitted
Tuesday	+0.018	0.010	0.065	+1.8%	
Wednesday	+0.023	0.012	0.056	+2.4%	
Thursday	+0.029**	0.014	0.041	+2.9%	
Friday	−0.010	0.012	0.386	−1.0%	
Saturday	−0.135***	0.013	< 0.001	−12.6%	No displacement
Sunday	−0.302***	0.016	< 0.001	−26.0%	Main effect

Notes: Coefficients are Bergen $\times$ Day_{*d*} interactions from Equation (6), Monday omitted as baseline. Standard errors clustered at the county-year level. If temporal displacement to Saturday were occurring, the Saturday coefficient should be positive; instead it is significantly negative (−12.6%), ruling out the most plausible displacement channel. Midweek effects are economically negligible (2–3%) relative to the 26% Sunday main effect. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

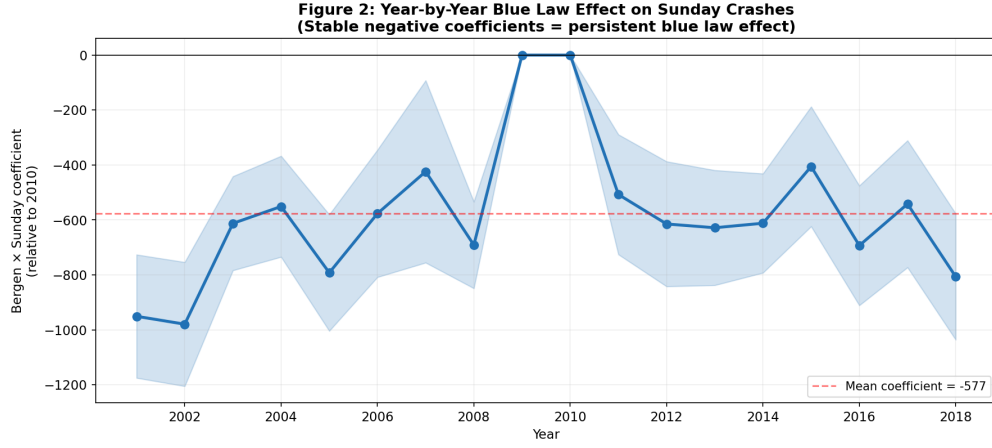
H Figure 1: Bergen County's Sunday Crash Deficit

Figure 1: Bergen County's Sunday Crash Deficit



Notes: Panel A plots the Sunday/Saturday crash ratio for each county, 2001–2018. Bergen (blue) persistently falls 10–15 percentage points below control counties. Panel B shows the day-of-week crash profile normalized to Monday = 1. Counties track closely Monday through Saturday, with Bergen already somewhat lower on Saturday (see Table 7); Bergen diverges sharply downward on Sunday.

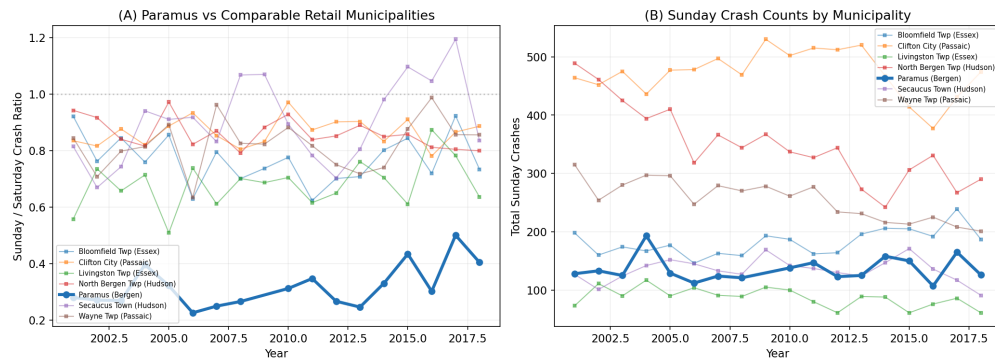
I Figure 2: Event Study – Year-by-Year Blue Law Effect



Notes: Each point shows the $\text{Bergen} \times \text{Sunday} \times \text{Year}$ interaction coefficient from Equation (3), with 2010 as the omitted reference year. Each $\hat{\delta}_y$ measures the blue law effect in year y relative to the 2010 treatment effect – not relative to zero. Shaded band is the 95% confidence interval. Persistently negative and stable coefficients confirm the blue law effect is a structural feature rather than a transitory shock.

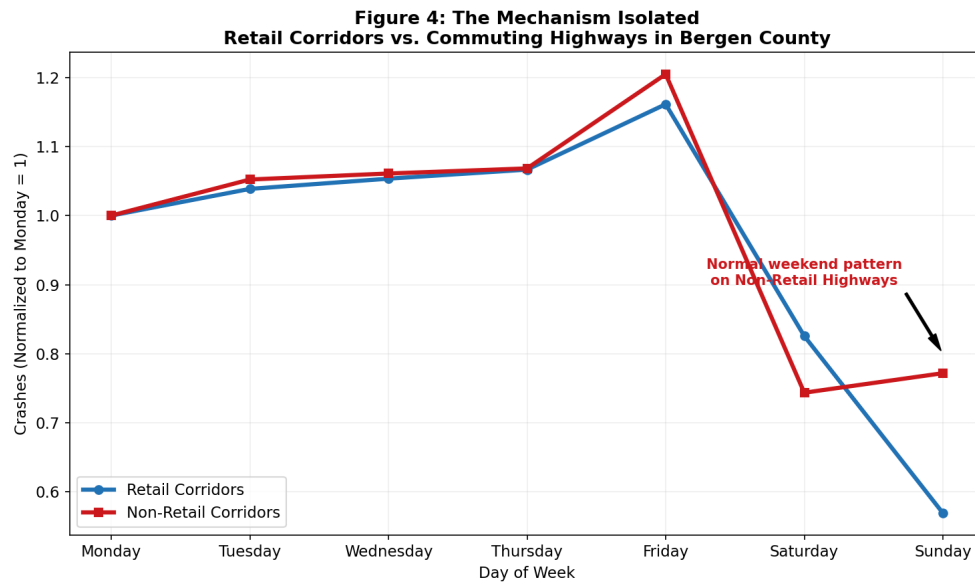
J Figure 3: Municipality-Level Analysis – Paramus vs. Comparable Retail Towns

Figure 3: Paramus Blue Law Effect – Municipality-Level Analysis



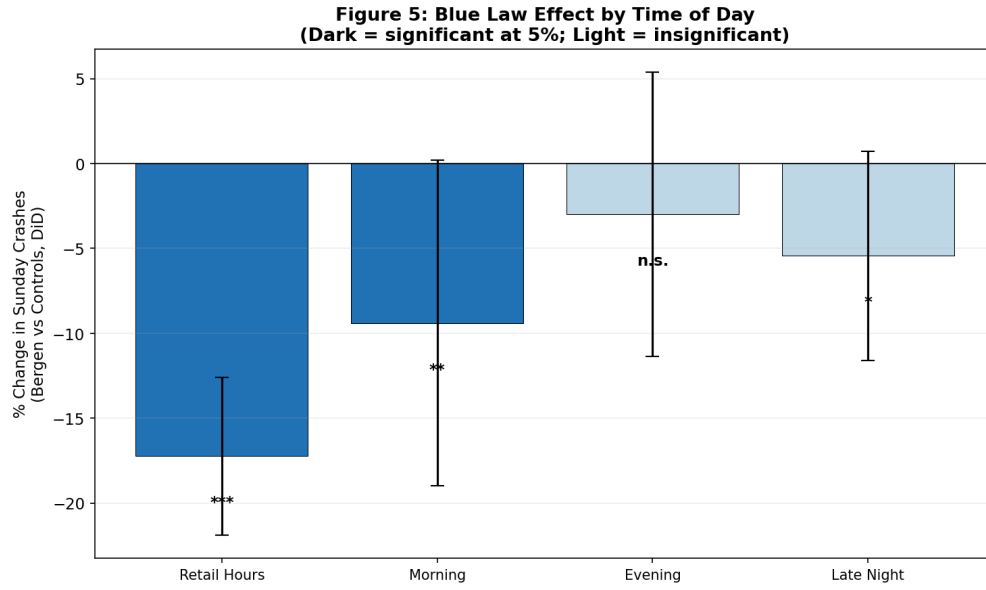
Notes: Panel A plots the Sunday/Saturday crash ratio for Paramus and comparable retail municipalities. Paramus's ratio (≈ 0.31) is dramatically lower than all comparables (0.67–0.89). Panel B shows Sunday crash counts in levels.

K Figure 4: Route-Level Mechanism – Retail vs. Commuter Highways
(Bergen County Only)



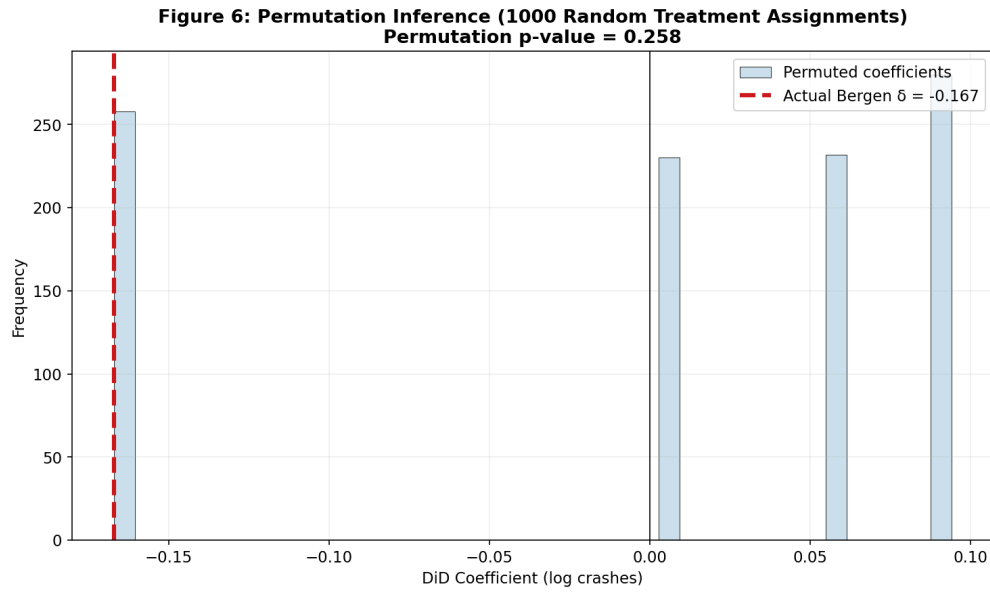
Notes: Crashes on Route 4 and Route 17 (retail corridors) compared with Garden State Parkway, I-95, I-80, and I-287 (non-retail highways), all within Bergen County. Retail corridors show a massive Sunday drop; non-retail highways show normal weekend patterns. This within-county comparison eliminates cross-county cultural confounds.

L Figure 5: Time-of-Day Heterogeneity



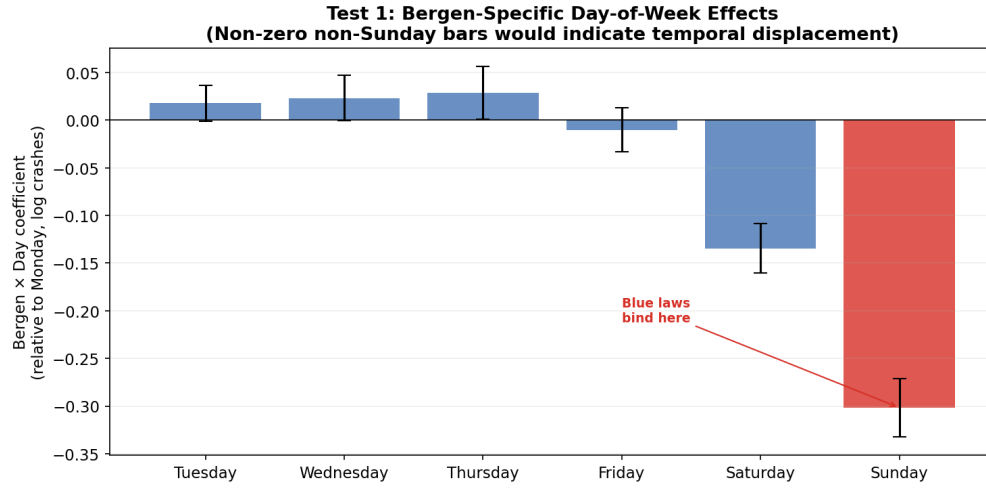
Notes: DiD estimates (log specification) run separately by time period. The blue law effect is concentrated during retail business hours (10am–9pm, -17.2%), with smaller but significant effects in the morning (-9.4% , $p = 0.085$) and late night (-5.4% , $p = 0.001$). The evening effect is small and statistically insignificant (-3.0% , $p = 0.426$). Dark bars indicate significance at the 5% level.

M Figure 6: Permutation Inference



Notes: Distribution of DiD coefficients from 1,000 random reassignments of treatment status across 4 counties. The red dashed line marks Bergen's actual coefficient (-0.167). Permutation $p = 0.258$, at the theoretical minimum given 4 clusters (≈ 0.25), indicating Bergen produces the most extreme Sunday deficit of any county. The permutation test circumvents the small-cluster problem in asymptotic inference.

N Figure 7: Temporal Displacement Test – Full-Week Bergen $\times$ Day Coefficients



Notes: Point estimates and 95% confidence intervals for Bergen $\times$ Day_{*d*} interaction coefficients from Equation (6), with Monday as baseline. Sunday shows the large blue law effect (−30.2%). Saturday is itself significantly negative (−12.6%), ruling out temporal displacement to the adjacent shopping day. Midweek coefficients are near zero. Standard errors clustered at the county-year level.